Warren "Chase" Carstensen IV

Machine Learning and Mathematics Researcher

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Education

Boston University **Boston, MA**

Master of Science in Artificial Intelligence *August 2026 – Present (Expected December 2027)*

Worcester Polytechnic Institute **Worcester, MA**

Dual Bachelor of Science in Computer Science and Mathematical Sciences, GPA: 3.93/4.0 *August 2022 – May 2026*

- **Selected Coursework**: Principles of Real Analysis I–II, Topology, Group Theory, Combinatorics, Graph Theory, Mathematical Statistics, Linear Programming, Algorithms, Numerical Methods
- **Extracurricular Activities**: WPI Jazz Groups

Research Experience

Major Qualifying Project **Worcester, MA**

Advisors: Prof. Daniel Reichman and Prof. Gábor Sárközy *August 2025 – March 2026*

- Project: *AI-in-the-Box: Generating Snake-in-the-Box Constructions with Neural Networks*
- Formulated the snake-in-the-box problem (finding long induced paths in hypercubes Q_d) as a reinforcement learning task.
- Developed the first deep learning-based approach to this problem.
- Reimplemented the Wagner framework (deep cross-entropy method) and PatternBoost (a transformer-based search method) to evaluate neural search strategies in higher dimensions.

CU Natural Language Processing REU **Colorado Springs, CO**

Advisor: Prof. Jugal Kalita *May 2025 – August 2025*

- Project: *Using Large Language Models to Generate Counterexamples in Graph Theory*
- Curated benchmark datasets of false graph-theoretic claims across 9 core domains (e.g., connectivity, spectral properties, extremal graph theory, forbidden minors).
- Benchmarked LLM counterexample generation against heuristic, genetic, and RL-guided search algorithms.
- Found that LLMs quickly find counterexamples to simple false claims (often faster than genetic or RL methods), but struggle on more complex graph properties.

WPI Data Science REU **Worcester, MA**

Advisor: Prof. Elke Rundensteiner *May 2024 – August 2024*

- Project: *MInDS: Using Large Language Models to Screen for Depression*
- Developed an automated zero-shot binary classification tool for depression screening from interview transcripts.
- Designed and implemented a greedy transcript paraphrasing algorithm to compress dialogue while preserving diagnostic features.
- Built an evaluation framework for benchmarking model screening capabilities, utilized by PhD students in the lab.

Publications

[1] Nathaniel Itty, Christopher D. Rosin, **Chase Carstensen**, and Daniel Reichman. "Improved lower bounds for the Shannon capacity of odd cycles." *arXiv preprint arXiv:2607.21517 [cs.IT]*, 2026.

[2] **Chase Carstensen** and Jugal Kalita. "Using Large Language Models to Generate Claims and Counterexamples in Graph Theory." In *Recent Advances in Natural Language Processing (REU Proceedings)*, 2025.

[3] **Chase Carstensen**, Nicholas Small, Janya Bhaskar, Becks Lopez, Avantika Shrestha, and Elke A. Rundensteiner. "MInDS: Using Large Language Models to Screen for Depression." In *2024 IEEE MIT Undergraduate Research Technology Conference (URTC)*, pp. 1–5, 2024.

Presentations

- **Using Large Language Models to Generate Counterexamples in Graph Theory**, VAST Lab, CU *(Colorado Springs, CO, August 2025)*
- **MInDS: Using Large Language Models to Screen for Depression**, IEEE MIT Undergraduate Research Technology Conference, MIT *(Cambridge, MA, October 2024)*
- **MInDS: Using Large Language Models to Screen for Depression**, DAISY Lab, WPI *(Worcester, MA, September 2024)*

Academic Experience

WPI Mathematical Sciences Department **Worcester, MA**
Peer Learning Assistant *August 2023 – May 2026*

- Facilitated weekly recitations and discussion sections for cohorts of ∼35 students across Calculus I–IV, Theoretical Calculus III–IV, and Graph Theory.
- Conducted weekly tutoring in the Math Tutoring Center, held dedicated office hours, and graded assignments and examinations.

Pi Mu Epsilon (National Mathematics Honor Society) **Worcester, MA**
President (Massachusetts Alpha Chapter) – Advisor: Prof. Brigitte Servatius *September 2024 – May 2026*

- Revitalized WPI's dormant chapter: rewrote the chapter constitution, restructured executive board governance, and streamlined candidate induction.
- Founded, organized, and hosted the inaugural annual WPI Undergraduate Mathematics Symposium.

Upsilon Pi Epsilon (International CS Honor Society) **Worcester, MA**
Events Coordinator (Massachusetts Alpha Chapter) – Advisor: Prof. Craig Wills *January 2025 – March 2026*

- Co-organized the annual honor society career fair and coordinated member volunteering across departmental initiatives.

Technical & Theoretical Skills

Programming Languages: Python, C++, Rust, C, Java, MATLAB, Bash, LaTeX
ML & Scientific Computing: PyTorch, Hugging Face Transformers, NetworkX, Pandas, NumPy, SciPy, Scikit-learn
Developer Tools & AI: Linux, Git, Slurm / HPC, Claude Code, Antigravity
Theoretical Specializations: Computational Combinatorics, Combinatorial Optimization, Graph Theory, Extremal Combinatorics, Reinforcement Learning

Honors & Awards

Meritorious Senior Award, *May 2026* – for outstanding academic performance in the Mathematical Sciences department

Honorable Mention, Provost's Award for Best MQP in Computer Science, *May 2026*

CRA Travel Grant, *October 2024* – to present at the IEEE MIT Undergraduate Research Technology Conference

High Distinction, Charles O. Thompson Scholar, & Dean's List (all semesters), WPI, *2022 – 2026*